

V1: Wellness Copilot for Colorectal Cancer Criteria

Overview

Beginning a pilot of the colorectal cancer gaps-in-care integration by the end of H2 2025 is a high priority. The goal is to start with an Epic OCHIN customer due to the availability of the data and the research already done to find where each data element lives in Epic Caboodle. We will want to expand this integration to each major EHR used by our customers.

I have discussed with Cecilia and Patrick and we have four separate messaging programs for the V1 version of the colorectal wellness copilot:

1. [Program #1 - Colorectal Cancer Prevention](#)
 - a. This program begins messaging patients at age 35 to begin educating on the importance of prevention for colorectal cancer.
 - b. This program can also be used in place of Program #2, if we do not have access to the clinical data needed to power Program #2.
2. [Program #2 - Colorectal Cancer Screening Selection](#)
 - a. This program utilizes the standard eCQI measures criteria to message patients who are due for screening.
3. [Program #3 - Reminders to complete and return at-home screening kits](#)
 - a. This program messages patients who have been ordered an at-home screening kit (FOBT/FIT/Cologuard) and reminds them to complete and kit and send it in
4. [Program #4 - Follow up from positive at-home screening kits](#)
 - a. This program reminds patients with positive FIT/FOBT/Cologuard kits to schedule their follow-up colonoscopy.

For future work planned, see [V2: Wellness Copilot for Colorectal Cancer Criteria](#)

References

[Priorities from Strategy Team](#)

[Deep Dive into Epic Caboodle Data for Colorectal Cancer](#)

[Colorectal Cancer Strategy - Canva](#)

Data Acquisition Strategy

We will need to implement these integrations for each EHR segment separately. It is important to prioritize these segments based off of ease of access and the size of the reachable patient population. Priority will also depend on how easy it is to secure pilot customers for each segment.

See the breakdown of where we currently are at with securing access to each data point per EHR here: [Data Availability](#)

1. Epic Caboodle OCHIN
 - a. Epic OCHIN is the #1 priority due to us already having access to the data and the fact that we have already researched the location and availability of the necessary data. We also have many OCHIN customers, so the reachable patient population is large.
2. Athena Health
 - a. Athena Health is another high priority due to the ease of access and reliability of their APIs and the high number of reachable patients.
3. eCW (with Ellkay)
 - a. eCW access with Ellkay is also a high priority. Although eCW bulk FHIR has proved to be unreliable thus far, we are currently undergoing validation with Ellkay. We need to validate that Ellkay will be able to provide us with lab orders and with referrals data - currently investigating.
4. Epic - non-OUCHIN
 - a. It's unclear how easy it will be to access the necessary data from non-OUCHIN Epic customers.
 - i. We will ask them to grant us access to Caboodle like OUCHIN did. If they say no to this, we could also ask them to set up extracts of their Clarity database.
 - ii. We are also undergoing validation to gain access to their FHIR APIs. We are just unsure how well this will scale due to the fact that Epic limits bulk FHIR to static groups of 1000 patients.
 - b. Avara is a potential pilot customer
5. Nextgen
 - a. We should be able to access the data that we need through a combination of the Nextgen proprietary APIs and FHIR APIs. We have already successfully built gaps-in-care integrations with Nextgen.
6. Practice Fusion
 - a. Practice Fusion does have bulk FHIR APIs available. It's unclear how much interest there is in this integration from our FCC Practice Fusion customers. They seem to be mainly interested in Appointment Reminders integrations for now, but perhaps we could bundle these integrations together (with Ellkay?)

Messaging Programs

I have written out four example messaging programs to demonstrate what could be achieved with the data that we have access to (and what data we think would be achievable). These are just suggestions and are pending input from strategy.

Goal #1: Prevention

Goal: Educate patients on screening options and early signs. The main goal of this intervention is to begin to educate patients on the importance of colorectal cancer screenings. The plan is to suggest this intervention be sent once annually, during CRC Awareness in March.

Numerator:

If using Goal #2 (Screening Selection):

- Limit to patients ages 40-45, or as young as 35 as an early intervention.

If NOT using Goal #2 (Screening Selection):

- Patient ages 45-75, and can also apply early education as young as 35

Denominator:

1. Patients with a visit in a primary care department within the clinic in the last 2 years (time frame configurable by customer)
2. Patients who have NOT had a total colectomy
3. Patients who have NOT been diagnosed with colorectal cancer

Overview of Messaging:

Q40 Colorectal Cancer Education

Data Dependencies:

Per Cecilia, the only data dependency for this intervention is age, although we may expand messaging to target based on other demographics in the future.

~~Demographics data (race, ethnicity, sex)~~

~~Family medical history~~

Frequency:

Annually, can align with CRC Awareness month in March

Questions for Strategy:

~~Need input from strategy here. We are able to access demographics data (race, ethnicity, sex, age) across EHRs. Family History is likely achievable to obtain across our major EHRs (see [Data Availability](#) section below), although we have not yet validated access for any EHR today.~~

Per Cecilia, this intervention will primarily be based on age and will be used to provide annual preventative education on the importance of colorectal cancer screening.

Level of Effort: LOW

- Since this intervention will be based on only on age for now, this intervention will be very easy to implement.
- Our EHR integrations already update the patient date of birth in the CareMessage patient profile. We could even build this intervention with the dynamic cohorts functionality within CareMessage.

Program #2: Screening Selection

Goal: Find patients who are due for colorectal cancer screening utilizing the [eCQI criteria](#) and remind them to schedule their screening. For organizations where access to their data allows us to refine how we find patients due for a colorectal cancer screening.

Numerator:

1. Patients who meet all of the following criteria:
 - a. No record of FOBT/FIT test in the last 12 months
 - b. No record of Stool DNA (sDNA) with FIT test (Cologuard) in the last 3 years
 - c. No record of Flexible Sigmoidoscopy (Flex Sig) in the last 5 years
 - d. No record of CT Colonography in the last 5 years
 - e. No record of Colonoscopy in the last 10 years

Denominator:

1. Patients with a visit in a primary care department within the clinic in the last 2 years (time frame configurable by customer)
 - a. Needs input from strategy: I'm imagining we don't want to message patients who only had a visit in a dental clinic, optometry, chiropractic, etc.
2. Patient age 45-75
3. Patients who have NOT had a total colectomy
4. Patients who have NOT been diagnosed with colorectal cancer

Overview of Messaging:

Q41 Colorectal Cancer Screening Support - this program informs patients about all their screening options

OR

Outreach Message if the priority is additional language, but the interaction will be a single message and very limited

Data Dependencies:

Lab Results for FOBT/Cologuard screenings.

Referrals and Procedures data for colonoscopies, flexible sigmoidoscopy, and CT colonography (not reports, just whether or not procedure was done).

Procedure data for total colectomy exclusion.

Diagnosis data for colorectal cancer exclusion.

Frequency:

Logic runs monthly, but only message X times per year

Questions for Strategy:

~~Do we want to find patients who are OVERDUE for their screening? Or do we want to be more proactive and message patients who are almost DUE for their screening?~~

- Suggestion from Patrick: "A gentle nudge that it is upcoming within next 60 days and then a separate message for overdue."
- Per Cecilia: we will limit the cohorts to patients who are overdue

NOTE: the eCQI criteria states that hospice/palliative care should be an exclusion. We don't have this exclusion for AGIC-D currently - should this be added for all gaps in care interventions?

- **TODO:** Research a hospice/palliative care exclusion for all gaps in care programs, as Cecilia has provided input that these messages can be triggering to patients and families
 - Encounters with CPT codes 99377-99378 and also check for ICD-10: Z51.5 in conditions.
 - Per Patrick, this data will not be readily available in the EHR data. We will likely need customers to find in claims data. There could possibly be a feature in the core app that allows customers to upload a list of patients on hospice.

Level of Effort: MODERATE

- We have access (or the likely ability to gain access) to the necessary data for Epic, Athena, Nextgen, and eCW
- We could build this integration using our existing Rhapsody infrastructure

Program #3: At-Home Test, Return

Goal: Support patients in completing their at-home tests

Numerator:

- Patients who meet all of the following criteria
 - Had an order placed for a FIT kit or Cologuard in the last three months (time frame?)
 - Do NOT have a lab result for a FIT kit or Cologuard from after the date of the lab order

Denominator:

- All patients in service (the lab order for Fit/Cologuard will serve to filter patients correctly)

Overview of Messaging:

Q42 - Colorectal Cancer - FIT Kit Return

OR

Q43 - Colorectal Cancer - Cologuard Return

Data Dependencies:

Lab results from FOBT/Cologuard tests.

Orders for FOBT/Cologuard tests.

Frequency:

Logic runs monthly, but only message once per instance?

Questions for Strategy:

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Level of Effort: MODERATE

- We have access (or the likely ability to gain access) to the necessary data for Epic, Athena, Nextgen, and eCW
- We could build this integration using our existing Rhapsody infrastructure

Program #4: Follow up from positive non-invasive screenings

Goal: Whenever a patient has a non-invasive colorectal cancer screening done (including FOBT and Cologuard) if they test positive on this screening, they need to have a follow up colonoscopy. The goal of this program is to assist with follow up of patients with positive results and encourage them to schedule their colonoscopy. Clinic staff should be following up on these positive results shortly after they are received, but this could be augmented with our messaging to assist with follow up. We can also help follow up with older positive results that were never followed up on.

See: <https://nccrt.org/wp-content/uploads/2023/12/Block-F-16-Master-Ensuring-Follow-up-v4.pdf>

NOTE: Positive CT colonographies also need to be followed up on, but I found very few records of CT colonographies in the Epic Caboodle database (see [AGIC - Colorectal Cancer - CT Colonography](#)). Because of the high lift of obtaining this data and the overall rarity of the procedure, we recommend considering this procedure out of scope for follow up.

NOTE: This program should take precedence over Program #1. If the patient has a positive result to follow up on, their messaging should be more targeted.

Numerator:

1. Patients who meet all of the following criteria
 - a. Have had a positive (abnormal) result on a FOBT or Cologuard test result in the last 3 years (time frame?)
 - i. Per Cecilia - we can do a historical backfill once at integration (look back 3 years), but after this, we will just look back to positive results from the last month
 - b. Have no record of a successful referral for colonoscopy AFTER the date of the positive stool based screening result

Denominator:

All patients in service (the lab result for Fit/Cologuard will serve to filter patients correctly)

Overview of Messaging:

Q44 - Colorectal Cancer - Result Follow-Up

Data Dependencies:

Lab results from FOBT/Cologuard tests.

Referrals and Procedures data for colonoscopies (not reports, just whether or not procedure was done).

Frequency:

Logic runs monthly, but only message once per instance?

Questions for Strategy:

What is the timeframe of the look back period that we should check for positive screening results? Last 3 years? Any amount of time as long as the patient had a visit in the last 2 years (and no record of colonoscopy performed afterwards?)

Level of Effort: MODERATE

- We have access (or the likely ability to gain access) to the necessary data for Epic, Athena, Nextgen, and eCW
- We could build this integration using our existing Rhapsody infrastructure

Data Availability

	Epic OCHIN	Epic non OCHIN	Nextgen	Athena Health	eCW	Practice Fusion
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FOBT/FIT and Cologuard Lab Results	YES (validated) Through Caboodle LabComponentResultFact table	YES (pending validation) We should be able to access this data over the FHIR Observations endpoint (pending validation of performance). Alternatively, we may be able to secure Caboodle or Clarity data access.	YES (validated) Through chart/lab /results proprietary API we already have access to	YES (validated) Through bulk FHIR Observations (already have access to)	YES (pending validation) Technically, eCW has this data available over their bulk FHIR API, but the APIs are not reliable. We are currently in the process of validated the Ellkay solution for data access for eCW	YES (not validated yet) We could find these over the bulk FHIR Observation API. Or we could get through Ellkay API.
Orders for FOBT/FIT and Cologuard Tests	YES (validated) Available through Caboodle ProcedureEventFact table	YES (pending validation) This should be available via the ServiceRequest.Search (Order Procedure) (R4) FHIR API. We should be able to search this API by Patient FHIR ID and CPT code to find orders for FOBT and Cologuard.	YES (not validated yet) We should be able to use the chart/lab /orders proprietary API to find these orders	YES (not validated yet) We could either use the ServiceRequest FHIR API or the proprietary Order - Lab API	Maybe? We could potentially use the Ellkay FHIR ServiceRequest API? It's unclear if this FHIR API is actually going to be available for eCW with Ellkay and if it will actually contain lab results.	Possibly? Potentially with the Ellkay integration, the ServiceRequest FHIR API could be available? And hopefully that actually contains lab orders?
Colonoscopy, Flex Sig, or CT Colonography Referral Completed	YES (validated) Caboodle ReferralFact table and ProcedureEventFact tables	Maybe? For FHIR, we would likely need to search the DocumentReference API, as the ServiceRequest (Referral) API is not going to be useful. It's also possible we are able to gain Caboodle or Clarity access.	YES (validated) Through referral-orders proprietary API we already have access to There could also be more information in the medical-history API and procedures API	YES (validated) We have access to the Orders - Referrals API, where we can find referrals done for colonoscopies. Possibly there is data in the Procedure FHIR API?	Maybe? We need to verify that the ServiceRequest FHIR API through Ellkay contains referrals data. And we would need to add this API to our scope.	Maybe? If we were able to subsidize Ellkay for a Practice Fusion customer, we could likely fetch referrals from the ServiceRequest FHIR API.
Exclusion Criteria - Total Colectomy	YES (validated) Caboodle ProcedureEventFact table	YES (pending validation) We should be able to access this data over the FHIR Procedure endpoint (pending validation of performance). Alternatively, we may be able to secure Caboodle or Clarity data access.	YES (not validated yet) We should be able to find this either in the medical-history API and procedures API	YES (not validated yet) We should be able to find this in the Procedure FHIR resource	YES (pending validation) Should be able to find in GetProcedures API or in Procedure FHIR resource through Ellkay	YES (not validated yet) We could find these over the bulk FHIR Condition API. Or we could get through Ellkay API.
Exclusion Criteria - Colorectal Cancer	YES (validated) Caboodle DiagnosisEventFact table	YES (pending validation) We should be able to access this data over the FHIR Condition endpoint (pending validation of performance).	YES (not validated yet) We should be able to find this in either the chart/problems or chart/health-concerns /encounter-diagnosis proprietary APIs	YES (not validated yet) We should be able to find this in the Condition FHIR resource	YES (pending validation) Should be able to find in either GetProcedures, GetDiagnosis, or Condition FHIR resource through Ellkay	YES (not validated yet) We could find these over the bulk FHIR Procedure API. Or we could get through Ellkay API.
Demographics	YES (validated) Caboodle PatientDim table	YES (pending validation) We should be able to access this data over the FHIR Patient endpoint (pending validation of performance).	YES (validated) The persons proprietary API has this	YES (validated) Patient FHIR resource	YES (pending validation) This will be in the Patient FHIR resource through Ellkay	YES (not validated yet) This is available in the Patient FHIR resource. Or we could retrieve from Ellkay.